at power levels under about 1.5 kW unless there is danger due to possible ignition of explosive materials. At ranges between 1-5 kW, the availability of a robot in a particular coordinate system with specific characteristics or at a lower cost may determine the decision. Reliability of all types of robots made by reputable manufacturers is sufficiently good that this is not a major determining factor.

Example 3.5 Selection of a Motor

Simple mathematical calculations are needed to determine the torque, velocity, and power characteristics of an actuator or a motor for different applications. Torque is defined in terms of force times distance or moment. A force, f, at distance, a, from the center of rotation has a moment or torque, τ, i.e., $\tau = fa$. In general terms, power, P, transmitted in a drive shaft is determined by the torque, τ, multiplied by the angular velocity, ω. Power P is expressed as, $P = \tau\,\omega$. For an example, a calculation can tell one what kilowatt or horsepower is required in a motor used to drive a 2-meter robot arm lifting a 25 kg mass at 10 rpm. If the mass of the arm is assumed zero then, $P = (25 \times 9.81 \times 2) \times (2\pi \times 10/60) = 0.513$ kW. The use of simple equations of this type is often sufficient to make a useful approximation of a needed value. More detailed calculations can take place using the equations of statics and dynamics that apply.

3.5 GRIPPERS

Grippers are end-effectors, as introduced in Section 2.1.1, which are used to grasp an object or a tool, e.g., a grinder, and hold it. Tasks required by the grippers are to hold workpieces and load/unload from/to a machine or conveyer. Grippers can be mechanical in nature using a combination of mechanisms driven by electric, hydraulic, or pneumatic powers, as explained in earlier sections. Grippers can be classified based on the principle of grasping mechanism. For example, grippers can hold with the help of suction cups, magnets, or by other means. A gripper is then accordingly referred to as *pneumatic gripper*, *magnetic gripper*, etc. Another way to classify a gripper is based on how it holds an object, i.e., based on grasping the object on its exterior (*external gripper*) or interior (*internal gripper*) surface.

3.5.1 Mechanical Grippers

As shown in Fig. 2.4(a), mechanical grippers have their jaw movements through pivoting or translational motion using a transmission element, e.g., linkages or gears, etc. This is illustrated in Fig. 3.27. The gripper can be of single or double type. While the former has only one gripping device at the robot's wrist, the latter type has two. The double grippers can be actuated independently and are especially useful in machine loading and unloading. As illustrated in Groover et al. (2012), suppose a particular job calls for a raw part to be loaded from a conveyor onto a machine and the finished part to be unloaded onto another conveyor. With a single gripper, the robot would have to unload the finished part before picking up the raw part. This would consume valuable time in the production cycle because the machine would remain idle during these handling motions. With a double gripper, the robot can pick up the part from the incoming conveyor with one of its gripping devices and have it ready to exchange for the finished part on the machine. When the machine cycle is completed, the robot can reach in for the finished part with the available grasping device, and insert the raw part into the machine with the other grasping device. The amount of time spent in exchanging the parts or the time to keep the machine idle is minimized.

A gripper uses its fingers or jaws to hold an object, as illustrated in Fig. 3.28. The function of a gripper mechanism is to translate some form of power input, be it electric, hydraulic or pneumatic, into the grasping action of the fingers against the part. Note that there are two ways a gripper can hold an object, i.e., either by physical constriction as shown in Fig. 3.28(a) or by friction, as demonstrated in Fig. 3.38(b). In the former case, contacting surfaces of the fingers are made of approximately the same shape of the part geometry, while in the latter case the fingers must apply sufficient force to retain the part against gravity or accelerations. The friction method of holding a part is less complex and hence less expensive. However, they have to be designed properly with its surfaces having sufficient coefficient of friction so that the parts do not slip during motion. Example 3.6 illustrates the situation.

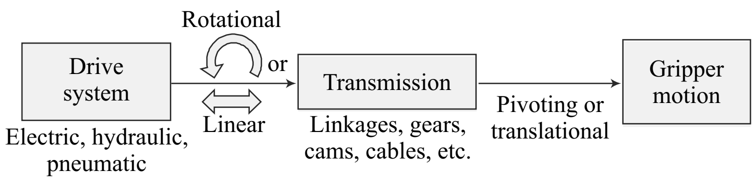


Fig. 3.27 Motion of a mechanical gripper

Example 3.6 Friction-based Gripper

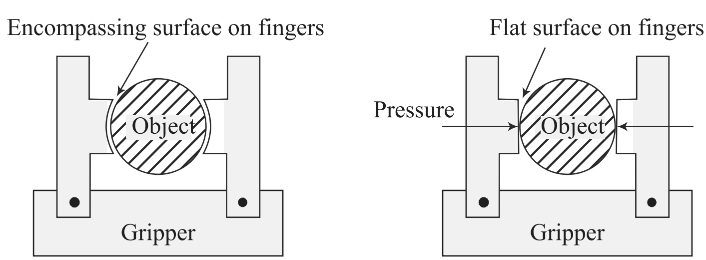
Consider the weight w of an object to be carried by a parallel-fingered gripper shown in Fig. 3.28(b). Gripper force can be calculated using the following force balance:

$$\mu n f = w \tag{3.5a}$$

where μ is the coefficient of friction of the object and finger surfaces, whereas n, f, and w are the number of contacting surfaces, finger forces, and the weight of the object to be held by the gripper. If w = 140 kg, n = 2, and μ = 0.2, f is computed simply as

$$f = \frac{140 \times 9.81}{2 \times 0.2} \cong 3500 \text{ N} \tag{3.5b}$$

Note that the value of 9.81 m/s^2 is the value of g, i.e., acceleration due to gravity.



(a) Physical or encompassing (b) Friction load

Fig. 3.28 Fingers gripping objects

Example 3.7 Gripper Accelerating with an Object

Assume the object of Example 3.6 is to be lifted up with an acceleration of 10 m/s^2. The gripper will require almost twice the force as the acceleration to be experienced by it is $(9.81 + 10) = 19.81 \cong 2g$. Rest of the calculation is same as in equation (3.5a-b).

Note that the motion of the fingers in mechanical grippers is obtained by means of any of the following transmission elements: (i) Linkage; (ii) Gear and rack; (iii) Cam; (iv) Screw; and (v) Cable and pulley, etc. Figure 3.29 shows some of these arrangements, where the transmission elements are either driven by an electric motor or a hydraulic/ pneumatic actuator. The final gripping action of the fingers is, however, achieved by one of the following means:

1. Pivoting Movement In this arrangement, the fingers rotate about fixed pivot points on the gripper to open and close. This is indicated in Fig. 3.29(a). The motion is usually achieved by some kind of linkage mechanism.

2. Translational Movement In the translational or linear motion, the fingers open and close by moving parallel to each other, as shown in Fig. 3.29 (b-d).

It is interesting to note that in the literature, e.g., in Groover et al. (2012), Deb and Deb (2010), and others, the mechanical grippers were classified as per either the transmission elements used or the type of finger movements, besides the input power, i.e., electric or pneumatic. For example, cam-operated gripper or pivoting gripper, etc. In the true sense, they should fall in the specifications of a mechanical gripper. Moreover, mechanical grippers with more fingers are also in use in industries, e.g., to hold spherical objects. One such robotic gripper, namely, a three-fingered hand, is shown in Fig. 2.4(b).

3.5.2 Magnetic Grippers

Unlike mechanical grippers, the principle of a magnetic gripper is based on the magnetic property of a gripper. Hence, they can be used only for ferrous objects. They have the following advantages:

- Variations in object sizes can be tolerated.
- Operations are very fast.
- Require only one surface to hold an object.

The disadvantages with magnetic grippers are, however, the difficulty to pick thin sheets one at a time because the magnetic force penetrates through more than one sheet. As a result, more than one sheet is picked up. To overcome such disadvantages, one needs to take care during the design stage itself either by limiting the magnetic force by the gripper or by introducing some means (mechanical or otherwise) not to allow more than one sheet to be picked up.

Magnetic grippers can have either (i) permanent magnets, or (ii) electromagnets. While electromagnetic grippers are easy to control requiring only a dc power source, grippers with permanent magnets do not require any external power source to operate the magnets. Besides, no electric sparks in handling hazardous materials. They, however, require an external stripping mechanism during the release of the object. One such mechanism is illustrated in Fig. 3.30.

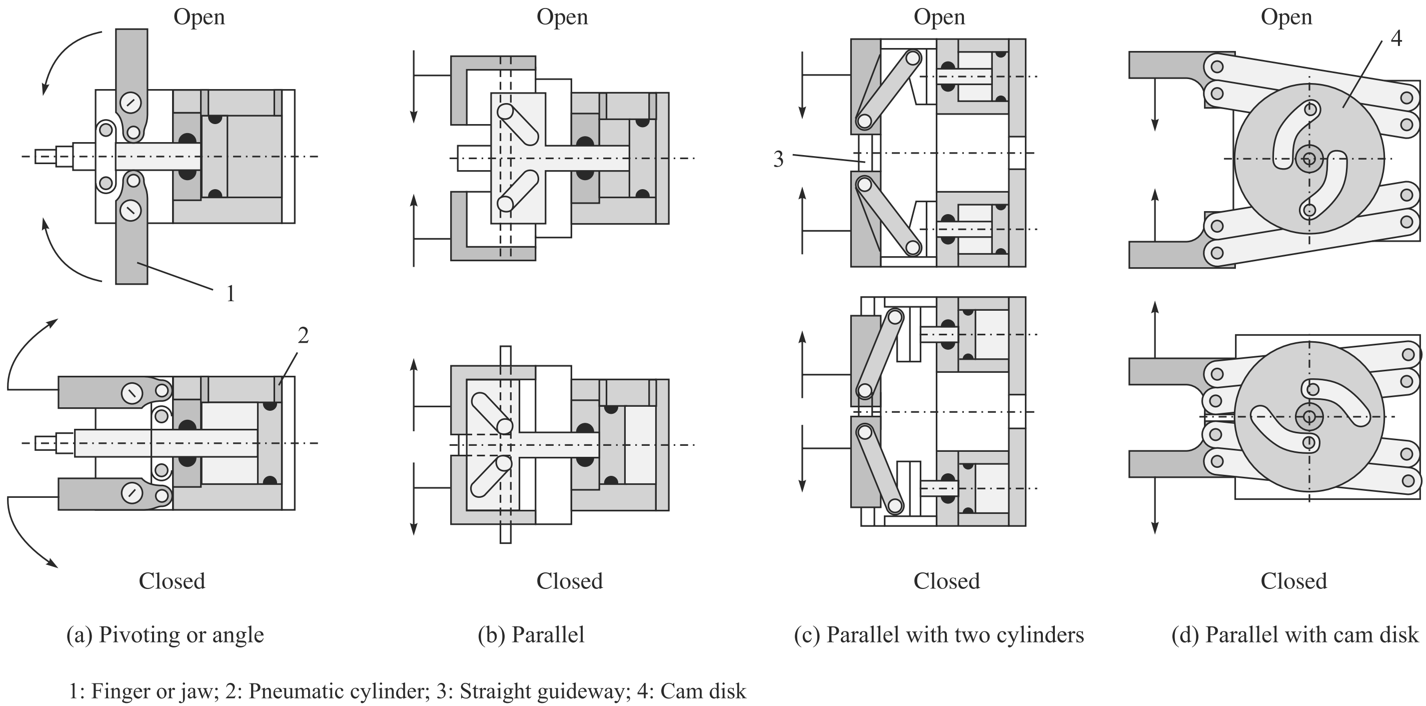


Fig. 3.29 Pivoting and translating mechanical grippers

[Courtesy: http://media.wiley.com/product_data/excerpt/90/35274061/3527406190.pdf]

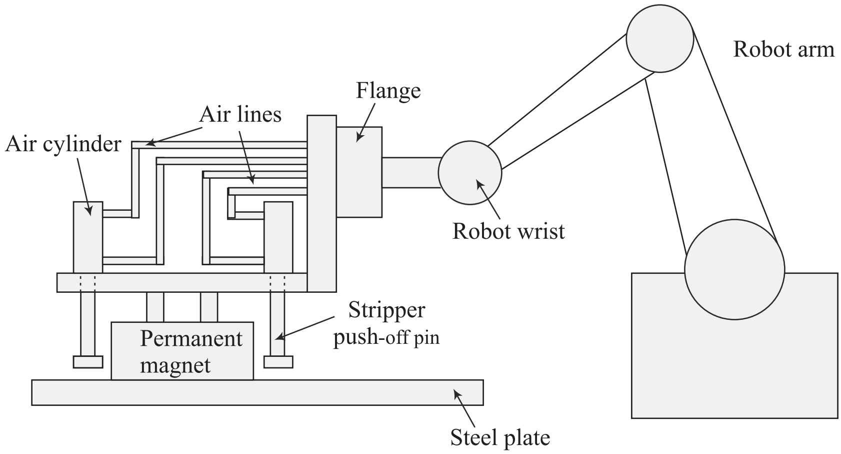


Fig. 3.30 Stripping of a sheet from a magnetic gripper

3.5.3 Vacuum Grippers

Such grippers are suitable to handle large flat objects. The material of an object is of no concern with vacuum gripers, except that the object's surface should not have any holes. An example of vacuum gripper which uses suction cups made of elastic materials is shown in Fig. 3.31. For a vacuum gripper, lifting capacity can be determined from the negative pressure and the effective area of the cups as

$$f = pA \tag{3.6}$$

where f is the force or lift capacity, p is the negative pressure, and A is the total effective area of the suction cups.

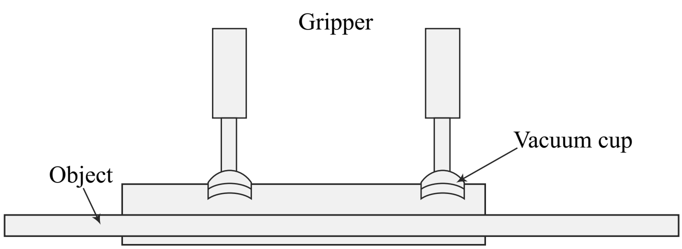


Fig. 3.31 Vacuum gripper

Example 3.8 Lifting Capacity

Consider lifting of four steel plates, each piece is 6 mm thick and measures 6 m × 10 m. The weight of the four steel plates is given by

$$w = 4 \times (0.006 \times 6 \times 10) \times 7800 \times 9.81 = 110187.12 \text{ N}$$

where 7800 kg/m^3 is the mass density of the steel plates, whereas 9.81 m/s^2 is the acceleration due to gravity. Assuming the diameter of vacuum cups as 100 mm, the required negative pressure can be obtained from Eq. (3.6) as

$$p = \frac{110187.12}{2 \times \pi \times (0.1)^2} = 175.35 \text{ N/m}^2 \tag{3.7}$$

3.5.4 Adhesive Grippers

An adhesive substance used for a grasping action can be used to handle fabrics and other lightweight materials. One of the limitations is that the adhesive substance loses its effectiveness with repeated use. Hence, it has to be continuously fed like a mechanical typewriter's ribbon which needs to be attached to the robot's wrist.

3.5.5 Hooks, Scoops, and Others

There exist other types of gripping devices, e.g., hooks, scoops or ladles, inflatable devices, etc., based on the need of item to be handled. For example, Fig. 3.32 shows one of these devices.

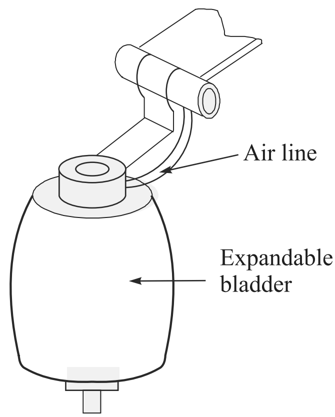


Fig. 3.32 Expandable bladder to grip objects with holes from inside

3.5.6 Selection of Grippers

Some of the criteria to be used in selecting an appropriate gripper are highlighted below:

- One needs to decide the drive system. In fact, in an industrial scenario where electric and pneumatic power sources are easily available, one needs to decide the type. If fast but not so accurate motion is desired then pneumatic should be preferred. Otherwise, one can go for electric. For heavy objects, certainly one must prefer hydraulic, as pointed in Section 3.2.
- Object or the part surface to be gripped must be reachable by a gripper.
- The size variation should be kept in mind. For example, a cast object may be difficult to put in a machine chuck.
- The gripper should tolerate some dimension change. For example, before and after machining a workpiece, their sizes change.
- Quality of surface area must be kept in mind. For example, a mechanical gripper may damage the surface area of an object.
- Force should be sufficient to hold an object and should not fly away while moving with certain accelerations. Accordingly, the materials on the surfaces of the fingers should be chosen.

SUMMARY

A robot is moved by actuators. Different forms of actuators, namely, electric, hydraulic and pneumatic types are explained. For electric motors, their typical specifications and speed-torque characteristics are presented. How to select a suitable motor is also outlined. Besides, grippers used to hold an object in order to transfer it to a new location are explained. Important aspects to select an appropriate gripper are also highlighted.